Iceberg FILE and VECTOR

- Proposal review for multimodal lakehouse architecture
- GitHub status checked October 23, 2026
- API, table semantics, and physical encoding have separate readiness boundaries

> 这组内容是设计提案与 PR，不是论文，也不是已稳定发版的能力。

FILE PR　Iceberg VECTOR　Parquet draft

Verified proposal status

Item	State on review date	Scope
Iceberg #18252	Open, not draft, not merged	FILE API and initial visitors
Iceberg #18297	Open issue	Fixed-size numeric vector proposal
Parquet #592	Open draft, not merged	VECTOR logical and repetition types

> API 响应核查显示三者均未形成稳定产品承诺。状态以后会变化，日期是本次快照。

FILE PR Iceberg VECTOR Parquet draft

FILE represents a byte range

- The proposed FILE type can reference bytes inline or in an external file.
- It defines a distinct nested type and TypeID instead of inheriting struct semantics.
- The submitted API change forms one part of a broader FILE support sequence.

> 读取图片或音视频片段需要字节范围的结构语义。API 类型存在并不代表所有 engine 的读写、GC 和权限已经实现。

FILE PR Iceberg VECTOR Parquet draft

Derived nested field IDs protect schema evolution

- Six nested fields derive their IDs from the containing FILE field ID.
- ID assignment reserves the derived block against future columns.
- Visitors receive dedicated FILE handling with a default struct fallback where appropriate.

> 类型系统必须防止 schema evolution 与内部字段 ID 冲突。普通 struct visitor 的兼容回退也需要逐个检查语义。

FILE PR Iceberg VECTOR Parquet draft

Version gating constrains rollout

- The FILE PR gates the proposed type to format version 4.
- Older table versions cannot silently acquire the new semantics.
- Reader, writer, schema evolution, and table upgrade support need independent verification.

> 版本门槛保护兼容性。不能因为格式规范拟新增一个类型就宣称现有 Spark/Trino 环境均可使用。

FILE PR Iceberg VECTOR Parquet draft

Iceberg VECTOR specifies shape intent

- The issue proposes a primitive fixed-size numeric vector type.
- Embedding dimension belongs in schema semantics rather than only application conventions.
- The current Iceberg issue points to Parquet proposal #624.

> 重要修正：给定阅读列表中的 Parquet #592 并非 issue 当前指向的唯一方案。不要把两个提案当成已经达成一致的最终规范。

FILE PR Iceberg VECTOR Parquet draft

Parquet #592 changes repetition semantics

- The draft adds a VECTOR logical type and VECTOR repetition type.
- SchemaElement.vector_length fixes the number of element slots per vector occurrence.
- The VECTOR repetition level adds no definition or repetition level of its own.

> 这是物理编码层面的新语义，不能把普通 LIST 写入后仅更名就当作兼容实现。

FILE PR Iceberg VECTOR Parquet draft

Nullability and reader behavior in #592

Aspect	Draft semantics
Nullable vector	Optional outer group
Nullable elements	Optional element field
Statistics and encodings	Remain element-level
Unsupported readers	Reject the VECTOR repetition type
Page boundary	Writers should keep one vector within a data page

> 这些是 #592 草案的具体选择，不保证最终标准采纳。兼容性测试需覆盖 null vector、null element、空值与页边界。

FILE PR Iceberg VECTOR Parquet draft

Prototypes provide limited performance evidence

- The draft links alternative Arrow prototypes and an Arrow Go proof of concept.
- Microbenchmarks motivate avoiding variable-list machinery for fixed shapes.
- No integrated Iceberg multi-engine production benchmark appears in these proposal bodies.

> 本 deck 不将提案链接中的微基准扩写为端到端训练吞吐增益，也没有执行任何原型。

FILE PR Iceberg VECTOR Parquet draft

Storage types leave indexing policy open

- Inference: a vector field still needs model version and distance-metric metadata.
- Inference: FILE references need lifecycle rules that protect external objects across snapshots.
- Proposed application: keep logical schema adoption separate from ANN index publication.

> 类型支持不自动提供 ANN、跨表事务或外部文件垃圾回收。以上是工程设计要求，不是 PR 已交付内容。

FILE PR Iceberg VECTOR Parquet draft

A compatibility-first evaluation matrix

- Test round trips across intended writers and readers at exact version pins.
- Test dimension mismatch, nulls, nested fields, field-ID evolution, and unsupported readers.
- For FILE, test overwrite, snapshot retention, external authorization, and orphan cleanup.

> 可以先在隔离样例数据上验证格式能力，避免升级真实训练表后才发现某个 engine 无法读取。

FILE PR Iceberg VECTOR Parquet draft

Adoption boundary

- Published state: FILE API work and VECTOR proposals remain open.
- Inference: the main risk lies in coordinated format and engine support.
- Proposed gate: wait for agreed spec and supported releases before production schema migration.

> 关注 Apache 评审、最终规范、release note 和跨引擎测试四条证据线。静态阅读与状态核查不等于集成验证。

FILE PR Iceberg VECTOR Parquet draft